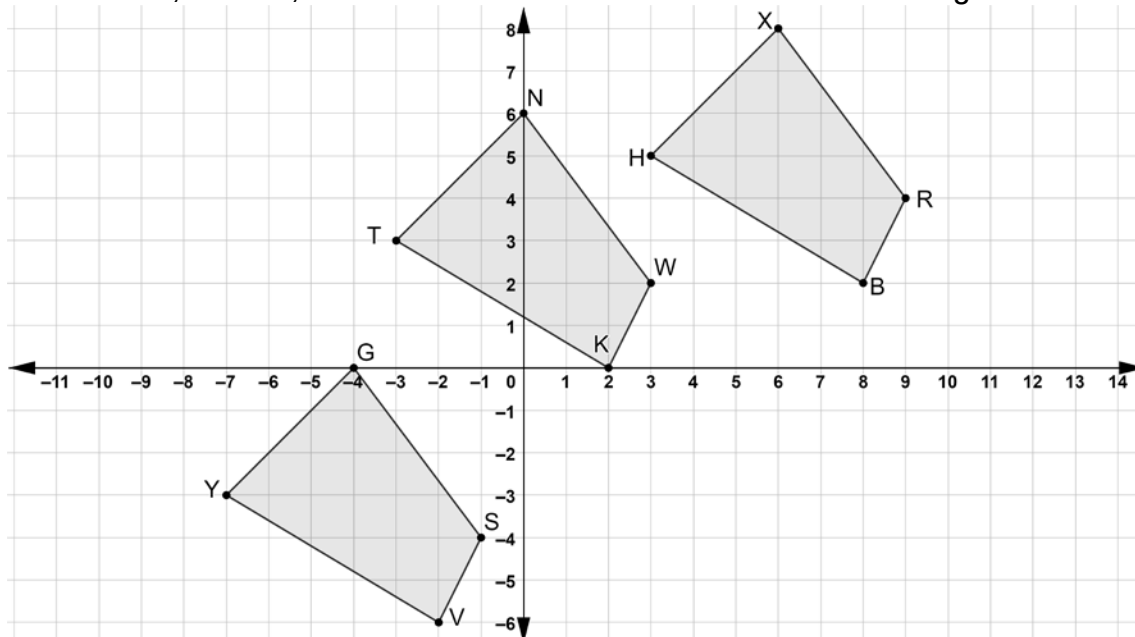


Quadrilaterals $GSVY$, $KWNT$, and $HBRX$ are shown on the coordinate grid.



1. Draw the mapping that shows the translation if quadrilateral $TKWN$ is the preimage and quadrilateral $HBRX$ is the image.
2. Describe the translation if quadrilateral $TKWN$ is the preimage and quadrilateral $HBRX$ is the image.
3. Complete the statements.
The x -values of quadrilateral $GSVY$, the image, can be found by

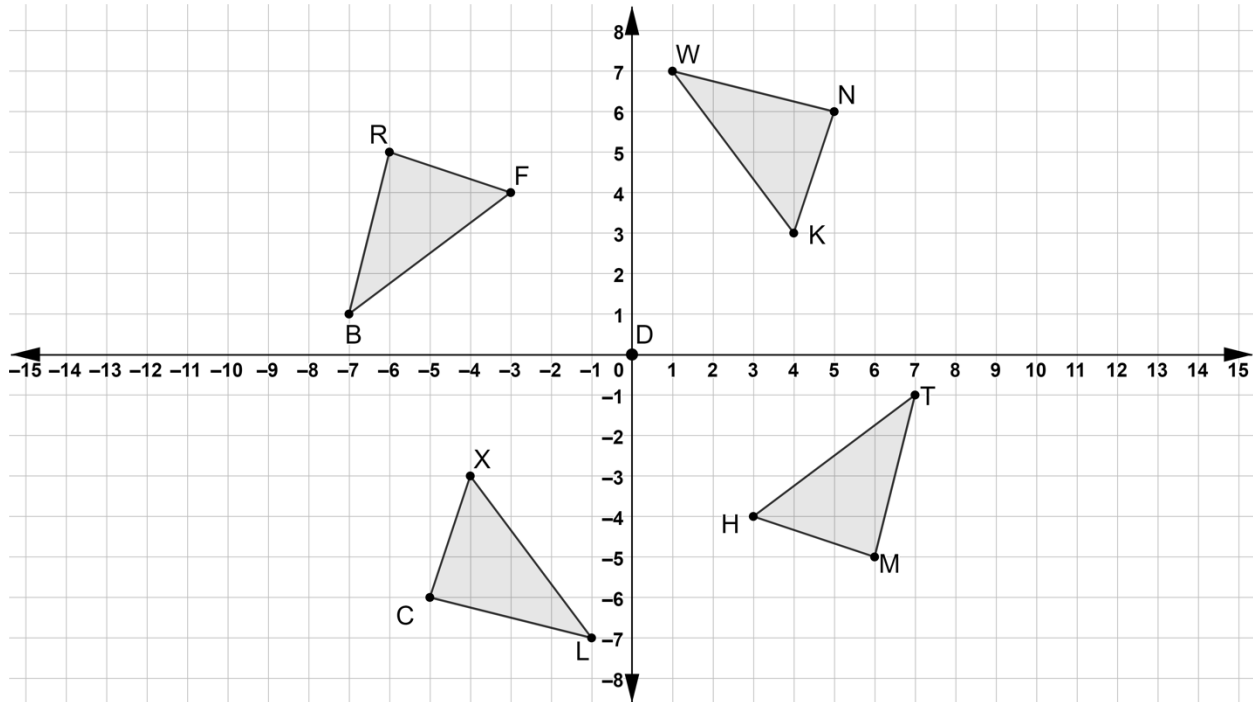
☐ adding
☐ subtracting
☐ multiplying

 _____ to the x -values of quadrilateral $XRHB$, the pre-image. The y -values of quadrilateral $GSVY$, the image, can be found by

☐ adding
☐ subtracting

 _____ units to the y -values of quadrilateral $XRHB$, the pre-image.
4. Describe the translation if quadrilateral $XRHB$ is the preimage and quadrilateral $GSVY$ is the image.

Four triangles are shown on the coordinate grid. Triangles WNK , TMH , and CXL are all rotations of $\triangle RFB$.



5. On the coordinate grid draw angle FDK .
6. Describe how the measure of $\angle FDK$ can be used to determine the degrees of rotation where $\triangle RFB$ is the pre-image and $\triangle NKW$ is the image.
7. Describe how the coordinates of the vertices of $\triangle RFB$, the preimage, and the coordinates of $\triangle NKW$, the image can be used to determine the degrees of rotation.
8. Describe the rotation where $\triangle RFB$ is the preimage and $\triangle CXL$ is the image.
9. Describe how the coordinates of the vertices of $\triangle RFB$ and $\triangle TMH$ show that rotating $\triangle RFB$ 180° about the origin results in $\triangle MHT$.
10. Explain why the rotation of $\triangle RFB$ that results in $\triangle MHT$ can be described as 180° clockwise or 180° counterclockwise.